Bol. Tema 3 y 4 : 2ª vez.

1- **X variable discreta**

Función masa probabilidad

$X \to P(X = x)$ ⟶

X variable continua

Función densidad f

$X \to F(x) = \int_{-\infty}^{x} f(t)\,dt$

Función distribución

$P(X \leq x)$

Acumulados.

Lanzamos 2 dados

a) X = menor valor observado

$d_1 \backslash d_2$	~~0~~	1	2	3	4	5	6
~~0~~	0	0	0	0	0	0	0
1	0	1	1	1	1	1	1
2	0	1	2	2	2	2	2
3	0	1	2	3	3	3	3
4	0	1	2	3	4	4	4
5	0	1	2	3	4	5	5
6	0	1	2	3	4	5	1

Masa Probabilidad

~~$P(X=0) = \frac{13}{36}$~~

$P(X=1) = \frac{11}{36}$

$P(X=2) = \frac{9}{36}$

$P(X=3) = \frac{7}{36}$

$P(X=4) = \frac{5}{36}$

$P(X=5) = \frac{3}{36}$

$P(X=6) = \frac{1}{36}$

b) F dist

$$P(X \leq x) = \begin{cases} \frac{11}{36} & \text{si } x < 2 \\ \frac{20}{36} & \text{si } x < 3 \\ \frac{27}{36} & \text{si } x < 4 \\ \frac{32}{36} & \text{si } x < 5 \\ \frac{35}{36} & \text{si } x < 6 \\ 1 & \text{si } x < 7 \end{cases}$$

c) $E[x] = X_i \cdot f_{rel_i}$

$= 1 \cdot \frac{11}{36} + 2 \cdot \frac{9}{36} + 3 \cdot \frac{7}{36} + 4 \cdot \frac{5}{36} + 5 \cdot \frac{3}{36} + 6 \cdot \frac{1}{36}$

$= 2'52\overline{7}$

$Var[x] = (X_i - E[x])^2 \cdot f_{rel_i}$

$= (1 - 2'52\overline{7})^2 \cdot \frac{11}{36} + (2 - 2'52\overline{7})^2 \cdot \frac{9}{36} + (3 - 2'52\overline{7})^2 \cdot \frac{7}{36}$

$+ (4 - 2'52\overline{7})^2 \cdot \frac{5}{36} + (5 - 2'52\overline{7})^2 \cdot \frac{3}{36} + (6 - 2'52\overline{7})^2 \cdot \frac{1}{36}$

$= 1'97$

d) $X =$ Par

$P(x=2) + P(x=4) + P(x=6) = \frac{9}{36} + \frac{5}{36} + \frac{1}{36} = \frac{15}{36}$

e) $P(x \leq 4) = 1 - P(x \geq 5) = 1 - P(x=6) - P(x=5)$

$= \frac{32}{36}$

$P(3 < x \leq 5) = P(x \leq 5) - P(x < 3) = P(x=5) + P(x=4)$

$= \frac{8}{36}$

2.

X_i	0	1	2	3	4
p_i	0'9	0'05	?	0'015	0'005

$x = 1 - 0'9 - 0'05 - 0'015 - 0'005$

$0'03$

a) $E[X] = \sum_{i=0}^{n} X_i \, p_i = 0 \cdot 0'9 + 1 \cdot 0'05 + 2 \cdot 0'03 + 3 \cdot 0'015 + 4 \cdot 0'005 = 0'175$

b) $Var[X] = (X_i - E[X])^2 p_i = 0'354375$

c) F dist $P(X \leq x)$
$$\begin{cases} 0'9 & si \; x < 1 \\ 0'95 & si \; x < 2 \\ 0'98 & si \; x < 3 \\ 0'995 & si \; x < 4 \\ 1 & si \; x < 5 \end{cases}$$

d) $P(X \leq 2) = 0'98$ $P(X > 3) = 1 - P(X \leq 2) = 0'02$

3. $X =$ " contestar bien una pregunta.

$X \sim Ber(p = 0'8)$

a) nº pruebas de Ber = 20 $X \sim Ber(n = 20, p = 0'8)$

$P(X = 10) = \binom{20}{10} 0'8^{10} \cdot 0'2^{10}$

b) _ _ _ E $X =$ " nº fracasos hasta primer éxito"

$X \sim G(p = 0'8)$

$P(X = 3) = q^x p = 0'2^3 \cdot 0'8$

c) X = " nº fracasos hasta el n éxito"

$X \sim BN(5, 0'8)$

_ _ E _ E _ E _ E _ E

$P(X=6) = \binom{x+n-1}{x} q^x p^n = \binom{10}{5} 0'2^6 \cdot 0'8^5$

fracasos

4- $p = \frac{5}{40} = 0'125$ de ser defectuoso.

$X \sim Bi(n=20, p=0'125)$

$P(X=2) = \binom{20}{2} 0'125^2 \cdot 0'875^{18} = 0'2683$

5- $p = 0'05$ $n = 10$

a) X = " nº piezas defectuosas en n tarjetas"

$X \sim Bi(10, 0'05)$

b) $P(X=1) = \binom{10}{1} 0'05^1 \cdot 0'95^9 = 0'315$

c) $P(X \geq 2) = 1 - P(X \leq 1) = 1 - \left(\binom{10}{1} 0'05^1 \cdot 0'95^9 - \binom{10}{0} 0'05^0 \cdot 0'95^{10} \right)$

$= 0'086$

d) X = " nº fracasos hasta primer éxito"

$X \sim G(p = 0'05)$ _ _ _ _ _ E

$P(X=5) = q^x p = 0'95^5 \, 0'05 = 0'038$

e) $X =$ " nº fracasos hasta el n éxito "

$X \sim BN(2, 0'05)$ — — — — — — — — E E

$P(X=7) = \binom{n+x-1}{x} q^x p^n = \binom{8}{7} 0'95^7 \cdot 0'05^2 = 0'037$

6: $X \sim Bi(n, p)$

$E[x] = np$ $\quad$ $Var[x] = n \cdot p \cdot q$

$20 = np$ $\quad$ $16 = np(1-p)$

$p = \frac{20}{n}$ $\quad$ $16 = 20 - \frac{400}{n} \Rightarrow -4 = \frac{400}{n} \Rightarrow n = 100$

$p = 0'2$

$X \sim Bi(100, 0'2)$ $\quad$ $P(X=25) = \binom{100}{25} 0'2^{25} \cdot 0'8^{75} = 0'0438$

7: $X =$ " nº de personas que suben al bus "

$Y_i =$ persona sube o no suble

$Y_i \sim Ber(p_i)$ $\quad$ $X = \sum_{i=1}^{n} Y_i$

$E[x] = E[\sum_{i=1}^{n} Y_i] = \sum_{i=1}^{n} p_i$

$Var[x] = Var[\sum_{i=1}^{n} Y_i] = \sum_{i=1}^{n} p_i q_i$

8.- 6 programas → 1 minuto.

x → 45 s

$$x = \frac{45 \cdot 6}{60} = 4'5 \text{ prog}$$

$X =$ "nº de sucesos en un intervalo de tiempo" $X \sim Pois(\lambda = 4'5)$

a) $P(X > 3) = 1 - P(X \leq 2) = 1 - \frac{e^{-4'5} \cdot 4'5^2}{2!} - \frac{e^{-4'5} \cdot 4'5^1}{1!}$

$- \frac{e^{-4'5} \cdot 4'5^0}{0!}$

b) 6 prog → 1 minuto $\quad x = \frac{15 \cdot 6}{60} = 1'5 \quad X \sim Pois(\lambda = 1'5)$

x → 15 s

$P(X \geq 1) = 1 - P(X \leq 0) = 1 - \frac{e^{-1'5} \cdot 1'5^0}{0!}$

9.- $n = 140 \quad p = 0'15$

a) $X \sim Bi(140, 0'15)$

$E[X] = 140 \cdot 0'15 = 21$

b) $P(X=21) = \binom{140}{21} \cdot 0'15^{21} \cdot 0'85^{119} = 0'094$

c) $P(X < 28) \quad Bi(140, 0'15) \sim N(np, npq)$

$N(140 \cdot 0'15, 140 \cdot 0'15 \cdot 0'85) \sim N(21, 17'85)$

$P(X \leq 27)$

27

$P(X \leq 27'5) = P\left(Z \leq \frac{27'5 - 21}{\sqrt{17'85}}\right) = P(Z \leq 1'54)$

10.- $E[X] = E[Y] = E[X \cdot Y] = 1$

$Var[X] = Var[Y] = 1$

$W = bX + cY$

$E[W] = E[bX + cY] = bE[X] + cE[Y] = 1$

$b + c = 1$

$Var[W] = E[(W - E(W))^2]$

$= E[(bX + cY - b - c)^2] = E[(b(X-1) + c(Y-1))^2]$

$= E[b^2(X-1)^2 + 2bc(X-1)(Y-1) + c^2(Y-1)^2]$

$= E[b^2(X-1)^2] + E[2bc(X-1)(Y-1)] + E[c^2(Y-1)^2]$

$= b^2 + 2bc + c^2 = 1$

$b^2 + 2bc + c^2 = 1$

$b + c = 1 \Rightarrow c = 1 - b$

$b = 1$ y $c = 0$

o

$b = 0$ y $c = 1$

11.- $F(x) = \begin{cases} 0 & \text{si } x \leq -1 \\ Kx^2 & \text{si } -1 < x \leq 1 \\ 0 & \text{si } 1 < x \end{cases}$

a) F densidad

$$F(x) = \int_{-\infty}^{\infty} f \, dt = 1$$

$$1 = \int_{-\infty}^{-1} 0x \, dx + \int_{-1}^{1} Kx^2 \, dx + \int_{1}^{\infty} 0x \, dx$$

$$\Rightarrow 1 = \left.\frac{Kx^3}{3}\right]_{-1}^{1} \Rightarrow 1 = \frac{K}{3} + \frac{K}{3} \Rightarrow K = \frac{3}{2}$$

b) $P(-0'5 < X < 0'5) = P(X \leq 0'5) - P(X \leq -0'5)$

$$= \frac{0'5^3 + 1}{2} - \frac{-0'5^3 + 1}{2} = 0'125$$

$$P(-2 < X \leq 0) = P(X \leq 0) - \underset{\substack{\downarrow \\ 0}}{P(X \leq -2)} = \frac{0^3 + 1}{2} = 0'5$$

c) F dist

$$F(X \leq x) \begin{cases} 0 & \text{si } x \leq -1 \\ \frac{x^3 + 1}{2} & \text{si } x \in [-1, 1] \\ 1 & \text{si } x > 1 \end{cases}$$

si $x \leq -1$

$$\int_{-\infty}^{x} 0 \, dt$$

si $x \leq 1$

$$\int_{-\infty}^{-1} 0 \, dt + \int_{-1}^{x} \frac{3}{2} t^2 = \frac{3}{2}\left(\frac{x^3}{3} + \frac{1}{3}\right)$$

12. $f(x) = \begin{cases} 0 & si \quad x < 1 \\ K(x-1)(x-3) & si \quad 1 \leq x \leq 3 \\ 0 & si \quad x > 3 \end{cases}$

$$1 = \int_1^3 K(x-1)(x-3)\,dx \Rightarrow 1 = K \int_1^3 x^2 - 4x + 3\,dx$$

$$\Rightarrow 1 = K \cdot \left[\frac{x^3}{3} - 2x^2 + 3x\right]_1^3 \Rightarrow 1 = K\left[9 - 18 + 9 - \frac{1}{3} + 2 - 3\right] \Rightarrow K = \frac{-3}{4}$$

F dist $\begin{cases} 0 & si \quad x < 1 \\ \frac{-x^3 + 6x^2 - 9x + 4}{4} & si \quad x \in [1,3] \\ 1 & si \quad x > 3 \end{cases}$

si $x < 1$

$$\int_{-\infty}^x 0\,dt = 0$$

si $1 \leq x \leq 3$

$$\int_{-\infty}^1 0\,dt + \int_1^x -\frac{3}{4}(t-1)(t-3)\,dt$$

$$= -\frac{3}{4}\int_1^x t^2 - 4t + 3 = -\frac{3}{4}\left[\frac{x^3}{3} - 2x^2 + 3x - \frac{1}{3} + 2 - 3\right]$$

$$= -\frac{3}{4}\left[\frac{x^3 - 6x^2 + 9x - 4}{3}\right] = \frac{-x^3 + 6x^2 - 9x + 4}{4}$$

$P(1'9 < x < 2'1) = P(x < 2'1) - P(x < 1'9) = 0'1445$

13- a) $P(x \leq 25)$ $\qquad N(20,9)$

$z = \frac{x-20}{\sqrt{9}}$ $\qquad P(z \leq \frac{5}{3})$

b) $P(x>5)$ $\qquad N(4,4)$

$z = \frac{x-4}{\sqrt{4}}$ $\qquad P(z > \frac{1}{2}) = 1 - P(z \leq \frac{1}{2})$

c) $P(x<70)$ $\qquad N(70,6)$

$z = \frac{x-70}{\sqrt{6}}$ $\qquad P(z<0)$

d) $P(13<x<17)$ $\qquad N(15,4)$

$z = \frac{x-15}{\sqrt{4}}$ $\qquad P(-1<z<1)$

$= P(z<1) - P(-1<z) = p(z<1) - (1 - p(1>z))$

$= 2P(z<1) - 1$

e) $P(|x|<3)$ $\qquad N(4,1)$

$z = \frac{x-4}{\sqrt{1}}$ $\qquad P(-3<x<3) = P(-7<z<-1)$

$= P(z<-1) - P(z<-7) = (1-P(z<1)) - (1-P(z<7))$

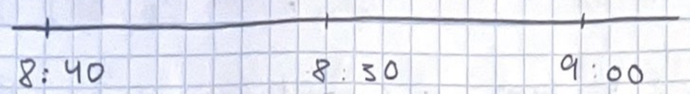
f) $P(|x|>7)$ $\qquad N(5,1)$

$z = \frac{x-5}{\sqrt{1}}$

$P(-7>x>7) = P(-13>z>2) = P(2<z<-13) = P(z<-13) - P(z<2)$

$= (1-P(z<13)) - P(z<2) = -P(z<2$

14-



X = " instante de llegar a clase " $X \sim U(a,b)$ 55, 65

a) $F(x) = \begin{cases} 0 & x < 55 \\ \dfrac{x-55}{65-55} & 55 \leq x < 65 \\ 1 & x \geq 65 \end{cases}$

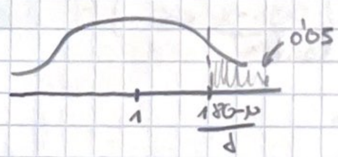
$P(X > 60) = 1 - P(X \leq 60) = 1 - \dfrac{60-55}{65-55} = \dfrac{1}{2}$

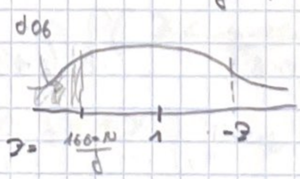
b) $E[X] = \dfrac{55+65}{2} = 60 \rightarrow$ 9:00

15- X = " medir cierta altura "

$P(X > 180) = 0'05$ $P(X < 160) = 0'06$

$P\left(Z > \dfrac{180-\mu}{\sigma}\right) = 0'05$ $P\left(Z < \dfrac{160-\mu}{\sigma}\right) = 0'06$





$1 - P\left(X \leq \dfrac{180-\mu}{\sigma}\right) = 0'05$ $P\left(Z < \dfrac{160-\mu}{\sigma}\right) = -0'94$

$P\left(X \leq \dfrac{180-\mu}{\sigma}\right) = 0'95$

$\left.\begin{array}{l} \dfrac{180-\mu}{\sigma} = 1'65 \\ \\ \dfrac{160-\mu}{\sigma} = -1'56 \end{array}\right\}$ $\mu = 169'7$ $\sigma = 6'27$ $N(169'7, 6'23^2)$

$P(x > 190) = P\left(z > \frac{190 - 169'7}{6'23}\right) = 1 - P(x \leq 3'25)$

$P(170 \leq x \leq 175) = P\left(\frac{170 - 169'7}{6'23} \leq z \leq \frac{175 - 169'7}{6'23}\right)$

$= P(x \leq 0'85) - P(x \leq 0'04)$

16- $E[x] = 8$ meses $X =$ "vida de la bombilla"

a) $P(3 < x < 12)$ $1 \longrightarrow 8$ meses

$X \sim Exp(\lambda)$ $E[x] = \frac{1}{\lambda}$

$= P(x \leq 12) - P(x \leq 3)$ $\lambda = \frac{1}{8}$

$= 1 - e^{-\frac{12}{8}} - 1 + e^{-\frac{3}{8}} = 0'4642$

b) $P(x > 25 \mid x > 10) = P(x > 15) = e^{-\frac{15}{8}} = 0'1534$

17- Máx 1200 $N(65, 225)$ 1 persona

si hay 20

Th límite $N(65 \cdot 20, 225 \cdot 20)$

$\sum_{i=1}^{n} X_i \sim (n\mu, n\sigma^2)$ $N(1300, 4500)$

cada persona

$P(x \geq 1200) = P\left(z \geq \frac{1200 - 1300}{\sqrt{4500}}\right) = P(z \geq -1'49)$

$= P(z \leq 1'49) = 0'9319$

18. $N(10, 3'5^2)$

X = "Periodo de garantía"

$P\left(Z < \frac{x-10}{3'5}\right) = 0'04$

$P\left(Z < \frac{x-10}{3'5}\right) = -1'75$

$\frac{x-10}{3'5} = -1'75$

$x = 3'875$ años

19. $N(100, 10^2)$

a) $P(90 < x < 115) = P(x < 115) - P(x < 90)$

$= P\left(Z < \frac{115-100}{10}\right) - P\left(Z < \frac{90-100}{10}\right)$

$= P(Z < 1'5) - P(Z < 1)$

b) $P\left(Z > \frac{x-100}{10}\right) = 0'1 \Rightarrow P\left(Z < \frac{x-100}{10}\right) = 0'9$

$\frac{x-100}{10} = 1'29 \qquad x_0 = 112'9$

c) $P(105 \leq x \leq c) = 0'1$

$P\left(0'5 \leq x \leq \frac{c-100}{10}\right) = 0'1$

$P\left(x \leq \frac{c-100}{10}\right) - P(x \leq 0'5) = 0'1$

$P\left(x \leq \frac{c-100}{10}\right) = 0'7915 \qquad \frac{c-100}{10} = 0'81 \qquad c = 108'1$

20: $3 \rightarrow 1$ min

a) X = "pase x tiempo hasta primer coche"

$X \sim Exp(\lambda)$ $\qquad \lambda = 3$

$P(X > 1) = 1 - (1 - e^{3})$

b) Y = "pase x tiempo hasta primer coche." $\quad Y \sim Pois(\lambda = 3)$

$$P(X \geq 1 \mid Y \geq 1) = \frac{P(X \geq 1 \cap Y \geq 1)}{P(Y \geq 1)} = P(X \geq 1) = e^{3}$$

c) Gamma.

d) $3 \rightarrow 1$ min $\qquad X$ = "nº de coches en 2 min"

$6 \rightarrow 2$ min $\qquad \lambda = 6 \qquad X \sim Pois(\lambda = 6)$

$P(X \geq 5) = 1 - (P(X = 4) + P(X = 3) + P(X = 2) + P(X = 1) + P(X = 0))$

e) $3 \rightarrow 1$ min $\qquad \lambda = 180$

$180 \rightarrow 60$ min $\qquad X$ = "nº de coche en 60 min"

$P(X > 200) \sim 1 - P(X \leq 200)$

$X \sim Pois(\lambda = 180) \sim N(180, 180)$

$$1 - P\left(Z \leq \frac{200'5 - 180}{\sqrt{180}}\right) = 1 - P(Z \leq 1'528)$$

$Z \leq 200$

21- 4 $\rightarrow$ 10 min

a) X = " probabilidad de x peticiones en 2 min "

$X \sim Pois\ (\lambda = 0'8)$

$4 \rightarrow 10$ min

$0'8 \rightarrow 2$ min

$P(X>0) = 1 - P(X \leq 1)$

$= 1 - \frac{e^{-0'8} \cdot 0'8^{1}}{1!} - \frac{e^{-0'8} \cdot 0'8^{0}}{0!}$

b) X = " probabilidad de esperar x tiempo entre 2 peticiones consecutivas "

$X \sim Exp(\lambda = 0'8)$

$P(X>2) = 1 - P(X \leq 1) = 1 - (1 - e^{0'8 \cdot 1})$

c) $4 \rightarrow 10$ mim

$12 \rightarrow 30$ min

X = " probabilidad de x peticiones en 30 min "

$X \sim Pois(\lambda = 12) \sim N(12, 12)$

$P(X>40) = P\left(Z > \frac{40'5 - 12}{\sqrt{12}}\right) = P(Z > 8'227)$


40

22 - $p = 0'3$

a) X = " tener fallos en n paginas " $X \sim Bi\ (n=5, p=0'3)$

$P(X=3) = \binom{5}{3} 0'3^{3} \cdot 0'2^{2}$

b) $X \sim Bi\ (n=50, p=0'3)$ $P(X \leq 21)$ $\mu = p$ $\sigma^2 = pq$

$X_1, \ldots, X_{50}$ Th central del limite $\sum_{i=1}^{n} X_i \sim N(n\mu, n\sigma^2)$

$N(50 \cdot 0'3,\ 50 \cdot 0'3 \cdot 0'7)$

$P(X \leq 21) = P\left(Z \leq \frac{21'5 - 15}{3'24}\right) = P(Z \leq 2)$


21

23- $Exp(\lambda)$ $E[X] = \frac{1}{\lambda}$ $\lambda = \frac{1}{2}$

a) $P(X<2) = 1-e^{-1}$

b) 50 clientes

$X_1, \dots, X_{50}$ th central del límite

$N(n\mu, n\sigma^2) \sim N(50 \cdot 2, 50 \cdot 4)$

$\mu = \frac{1}{\lambda} \Rightarrow \mu = 2$ $\sigma^2 = \frac{1}{\lambda^2} \Rightarrow \sigma^2 = 4$

$P(X<60) = P\left(Z < \frac{60-100}{\sqrt{200}}\right)$

24- $X \sim Pois(\lambda = 5)$ $5 \to 10$

a) $5 \to 10$ $Pois\ \lambda = 1$

$1 \to 2$

$P(X \geq 1) = 1 - P(X=0) = 1 - \frac{e^{-1} \cdot 1^0}{0!}$

b) $X \sim Exp(\lambda = 5)$

$P(X<0'5)$

c) $X_1, \dots, X_{50}$ $N(n\mu, n\sigma^2)$ $\mu = \frac{1}{\lambda} = \frac{1}{5}$

$\sigma^2 = \frac{1}{25}$

$P(X>180) = P\left(Z > \frac{180'5 - 0'2}{\sqrt{1/25}}\right)$

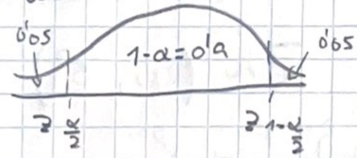
1- $N(\mu, s^2)$ $X_1, \dots, X_{16}$ $\bar{X} = 503$

a) $\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$ $\frac{\bar{X}-\mu}{\sigma} \cdot \sqrt{n} \sim N(0,1)$

$P\left(z_{\frac{\alpha}{2}} < \sqrt{n}\,\frac{\bar{X}-\mu}{\sigma} < z_{1-\frac{\alpha}{2}} \right) = 1-\alpha$

$P\left(\bar{X} - z_{1-\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}} , \bar{X} + z_{1-\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}} \right)$

$1-\alpha = 0'9$ $\alpha = 0'1$

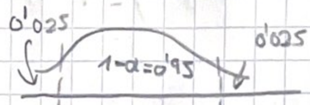


$-z_{0'95}$ $z_{0'95}$

$-1'65$ $1'65$

$IC_{0'9} = \left(503 - 1'65 \cdot \frac{5}{\sqrt{16}} , 503 + 1'65 \cdot \frac{5}{\sqrt{16}} \right) = (500'94, 505'0$

$IC_{0'95} = \left(503 - z_{1-\frac{\alpha}{2}} \cdot \frac{5}{\sqrt{16}} , 503 + z_{1-\frac{\alpha}{2}} \cdot \frac{5}{\sqrt{16}} \right)$

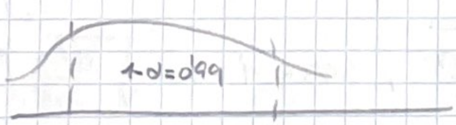


$-z_{0'975}$ $z_{0'975}$

$-1'96$ $1'96$

$IC_{0'95} = (500'55, 505'45)$

$IC_{0'99} = \left(503 - z_{1-\frac{\alpha}{2}} \cdot \frac{5}{\sqrt{16}} , 503 + z_{1-\frac{\alpha}{2}} \cdot \frac{5}{\sqrt{16}} \right)$



$-z_{0'995}$ $z_{0'995}$

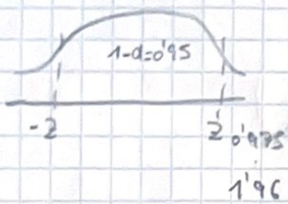
$-2'58$ $2'58$

$IC_{0'99} = (499'775, 506'225)$

b) $l = 1$ $\quad 1-\alpha = 0'95$ $\quad \alpha = 0'05$

$l = 2\, z_{1-\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}}$ $\quad n = \frac{4 z_{1-\alpha/2}^2\, \sigma^2}{l^2}$

$n = \frac{4 \cdot 1'96^2\ 5^2}{1^2} = 384'16$



c) $S_c^2 = 38'47$ $\quad \bar{X} \sim \left(\mu, \frac{\sigma^2}{n}\right)$

$\frac{(n-1) S_c^2}{\sigma^2} \sim \chi_{n-1}$

$\frac{\bar{X}}{\sqrt{\frac{Y}{n}}} \sim t_n$ $\quad \frac{\bar{X}-\mu}{\frac{S_c}{\sqrt{n}}} \sim t_{n-1}$

$P\left(t_{n-1, \frac{\alpha}{2}} < \frac{\bar{X}-\mu}{\frac{S_c}{\sqrt{n}}} < t_{n-1, 1-\frac{\alpha}{2}} \right) = 1-\alpha$

$P\left(\bar{X} - t_{n-1, 1-\frac{\alpha}{2}} \cdot \frac{S_c}{\sqrt{n}} ,\ \bar{X} + t_{n-1, 1-\frac{\alpha}{2}} \cdot \frac{S_c}{\sqrt{n}} \right)$

$1-\alpha = 0'1$

$t_{15, 0'95} \to 1'753$

$P\left(503 - 1'753 \cdot \sqrt{\frac{38'47}{16}} ,\ 503 - 1'753 \cdot \sqrt{\frac{38'47}{16}} \right)$

$I_{C\,0'9} = (500'28 ,\ 503'72)$

$I_{C_{0'95}}$ $(499'69 \quad , 506'30)$

$\alpha = 0'05$

$t_{15,\, 0'975} \to 2'131$

$I_{C_{0'99}}$ $(498'43,\ 507'57)$

$\alpha = 0'01$

$t_{15,\, 0'995} \to 2'947$

2- $X_1, \dots, X_{12}$ $\qquad S^2 = 0'00005 \qquad 1-\alpha = 0'99$

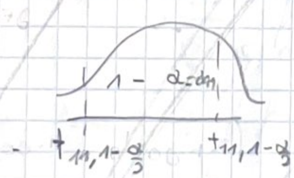
$$\frac{n S^2}{\sigma^2} \sim \chi^2_{n-1} \qquad \mu = 3\,cm$$

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right) - \sqrt{n-1}\ \frac{\bar{X}-\mu}{S} \sim t_{n-1}$$

$$P\left(t_{n-1,\frac{\alpha}{2}} < \sqrt{n-1}\ \frac{\bar{X}-\mu}{S} < t_{n-1,1-\frac{\alpha}{2}}\right) = 1-\alpha$$

$$I_C = \left(\bar{X} - t_{n-1,1-\frac{\alpha}{2}} \frac{S}{\sqrt{n-1}},\ \bar{X} + t_{n-1,1-\frac{\alpha}{2}} \frac{S}{\sqrt{n-1}}\right)$$

$$I_C = \left(3 - 3'106 \cdot \frac{\sqrt{0'00005}}{\sqrt{11}},\ 3 + 3'106 \cdot \frac{\sqrt{0'00005}}{\sqrt{11}}\right)$$



$\downarrow$

$t_{11,\, 0'995}$

$3'106$

$IC = (2'93,\ 3'07)$

3- $X_1, \dots, X_{50}$

$\hat{p} = 0'76$

$\hat{p} \sim N\left(p, \frac{pq}{n}\right)$

a) $1-\alpha = 0'95$

$\alpha = 0'05$

$\left(z_{\frac{\alpha}{2}} < \frac{\hat{p}-p}{\sqrt{\frac{pq}{n}}} < z_{1-\frac{\alpha}{2}} \right)$

solo en intervalos

$\left(\hat{p} - z_{1-\frac{\alpha}{2}} \sqrt{\frac{\hat{p}\hat{q}}{n}} , \hat{p} + z_{1-\frac{\alpha}{2}} \sqrt{\frac{\hat{p}\hat{q}}{n}} \right)$

$\left(0'76 - 1'96 \cdot \sqrt{\frac{0'76 \cdot 0'24}{50}} , 0'76 + 1'96 \sqrt{\frac{0'76 \cdot 0'24}{50}} \right)$

$z_{1-\frac{\alpha}{2}}$

$(0'64, 0'878)$

$z_{0'975} \to 1'96$

b) $l < 0'02$ $\quad l = 2z_{1-\frac{\alpha}{2}} \sqrt{\frac{p(1-p)}{n}}$

$$n = \frac{4 z_{1-\frac{\alpha}{2}}^2 \hat{p}(1-\hat{p})}{l^2} = \frac{4 \cdot 1'96^2 \cdot 0'76 \cdot 0'24}{0'02^2}$$

$n = 7007'2 \to 7008$

4: $p = 0'7$ $X_1, \dots X_{500}$ $\hat{p} = \frac{4}{5} = 0'8$

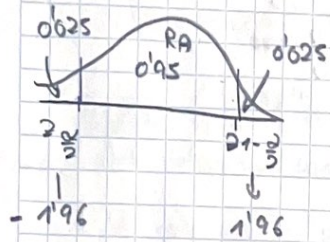
a) $\alpha = 0'05$

$H_0 : p = 0'7$

$H_1 : p \neq 0'7$

$\hat{p} \sim N\left(p, \frac{pq}{N}\right)$ $\qquad \dfrac{\hat{p} - p}{\sqrt{\frac{pq}{n}}} \sim N(0,1)$

$$\frac{0'8 - 0'7}{\sqrt{\frac{0'7 \cdot 0'3}{500}}} = 4'8795$$



$RA = (z_{\frac{\alpha}{2}}, z_{1-\frac{\alpha}{2}}) = (-1'96, 1'96)$

$RR = (-\infty, z_{\frac{\alpha}{2}}) \cup (z_{1-\frac{\alpha}{2}}, \infty)$

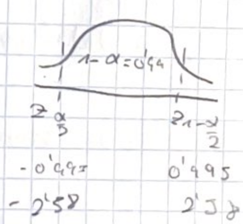
$= (-\infty, -1'96) \cup (1'96, \infty)$

como pertenece a la region de rechazo no puedo aceptar Ho.

b) $1 - \alpha = 0'99$ $\qquad \alpha = 0'01$

$$P\left(z_{\frac{\alpha}{2}} < \frac{\hat{p} - p}{\sqrt{\frac{pq}{n}}} < z_{1-\frac{\alpha}{2}}\right)$$

$$IC\left(\hat{p} - z_{1-\frac{\alpha}{2}} \cdot \sqrt{\frac{pq}{n}}, \hat{p} + z_{1-\frac{\alpha}{2}} \sqrt{\frac{pq}{n}}\right)$$



$$IC\left(0'7 - 2'58\sqrt{\frac{0'7 \cdot 0'3}{500}}, 0'7 + 2'58\sqrt{\frac{0'7 \cdot 0'3}{500}}\right)$$

$= (0'6472, 0'7528)$

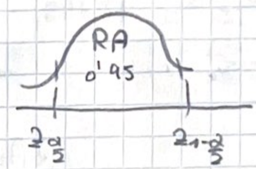
5- $\mu = 30 \quad \sigma = 5 \qquad X_1, \dots, X_{36} \qquad \bar{X} = 35$

a) $\alpha = 0'05 \qquad H_0: \mu = 30$

$\qquad\qquad\qquad H_1: \mu \neq 30$

$\bar{X} \sim \left(\mu, \frac{\sigma^2}{n}\right) \qquad \frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}} \sim N(0,1)$

$\frac{35-30}{\sqrt{\frac{25}{36}}} = 1642$



$z_{0'975}$

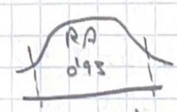
$-1'96 \qquad 1'96$

$RA = (-1'96, 1'96)$

$RR = (-\infty, -1'96] \cup (1'96, \infty)$ no podemos aceptar H_0 puesto que cae fuera de RA.

b) $S = 7 \qquad X_1, \dots, X_{36}$

$\sqrt{n-1}\,\frac{\bar{X} - \mu}{S} \sim t_{n-1} \qquad \sqrt{35} \cdot \frac{35-30}{7} = 4'226$



$t_{n-1, \frac{\alpha}{2}} \qquad t_{n-1, 1-\frac{\alpha}{2}}$

$-2'03 \qquad 2'03$

$RA = (-2'03, 2'03)$

$RR = (-\infty, -2'03) \cup (2'03, \infty)$

No podemos aceptarlo.

6.- $X_1, \dots, X_{20}$ $\quad s^2 = 0'005 \, cm^2$ $\quad \alpha = 0'1$

$\sigma^2 = 0'004$

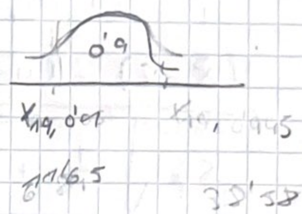
$$\frac{n \cdot s^2}{\sigma^2} \sim \chi^2_{n-1}$$

$H_0 : \sigma^2 \geq 0'004$

$H_1 : \sigma^2 < 0'004$

$$\frac{20 \cdot 0'005}{0'004} = 25$$

$$RA = \left(\chi_{n-1, \frac{\alpha}{2}}, \chi_{n-1, 1-\frac{\alpha}{2}} \right)$$



$RA = (11'65, \infty)$

$RR = (-\infty, 11'65)$

Sí que aceptamos H_0 puesto que cae en la RA

7.- $X_1, \dots, X_{100}$ $\quad p = 0'6$ $\quad X_1, \dots, X_{60}$ $\quad \bar{X} = 1'80 \, m$

$H_0 : \mu_X - \mu_Y \geq 0$ $\quad X_{60}, \dots, X_{100}$ $\quad \bar{Y} = 1'76$

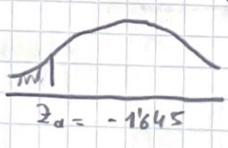
$H_1 : \mu_X - \mu_Y \leq 0$

a) $\sigma^2_x = 0'02$ $\quad \sigma^2_y = 0'015$ $\quad 1 - \alpha = 0'05$

$$\frac{(\bar{X} - \bar{Y}) - (\mu_x - \mu_y)}{\sqrt{\frac{\sigma^2_x}{n} + \frac{\sigma^2_y}{m}}} \sim N(0,1)$$

Suponiendo H_0:

$$\frac{1'80 - 1'76}{\sqrt{\frac{0'02}{60} + \frac{0'015}{40}}} = 1'502$$



$RR = (-\infty, -1'645)$

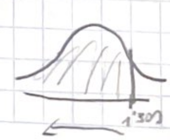
$RA = (-1'645, \infty)$ pertenece a la RA acepto $\mu_x \geq \mu_y$

p-valor $= P(z < 1'502) = 0'934$

$\alpha >$ p-valor $\rightarrow$ rechazaría H_0

$\alpha <$ p-valor $\rightarrow$ acepto H_0

$\alpha = 0'05 < 1'502$ acepto H_0.



2.- $n = 12 \qquad s^2 = 0'00003 \qquad \frac{ns^2}{\sigma^2} \sim \chi^2_{n-1}$

$$I_{(\sigma^2,\,1-\alpha)} = \left(\frac{ns^2}{\chi^2_{n-1,\,1-\frac{\alpha}{2}}} \;,\; \frac{ns^2}{\chi^2_{n-1,\,\alpha/2}} \right)$$

$99\% = 1-\alpha = 0'99$

$\alpha = 0'01$

$$I_{(\sigma,\,0'99)} = (0'0000021 \,,\, 0'00023)$$

t_{n+m-2}

7.- b) $$\frac{(\bar{x}-\bar{y}) - (\mu_x - \mu_y)}{S_T \sqrt{\frac{1}{n} + \frac{1}{m}}}$$

$$S_T = \sqrt{\frac{(n-1)\,S_{cx}^2 + (m-1)\,S_{cy}^2}{n+m-2}}$$

$$S_{cx}^2 = \frac{n}{n-1}\, S_x^2$$

$$S_T = \sqrt{\frac{(60-1)\cdot 0'006 + (40-1)\cdot 0'01}{40+60-2}}$$

$$S_{cx}^2 = \frac{60}{60-1} \cdot 0'08^2 = 0'0065$$

$$S_{cy}^2 = \frac{40}{40-1} \cdot 0'10^2 = 0'01$$

$S_T = 0'088$

$H_0: \mu_X \geq \mu_Y \rightarrow \mu_X - \mu_Y \geq 0$

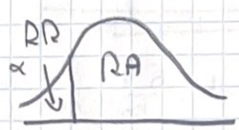
$H_1: \mu_X - \mu_Y < 0$

$$\frac{180-176}{0'087 \cdot \sqrt{\frac{1}{60} + \frac{1}{40}}} = 2'19$$

acepto H_0

$RA = (-1'660, \infty)$

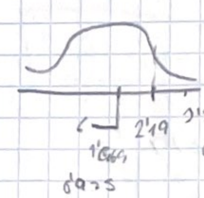
$RR = (-\infty, -1'660)$



$t_{n+m-2;\,\alpha}$ $\qquad \alpha = 0'05 \rightarrow 0'95$

$-1'660$

c) b)



p-valor $= P(T_{98} < 2'19)$

$0'975 < \text{p-valor} < 0'99$

$\alpha < 0'975 \rightarrow$ Acepto

$\alpha > 0'99$ rechazo